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Prediction of the next user location using personal tracking data

Sebastian Vallejos [a,b], Luis S. Berdun [a,b], Ariel J. Monteserin [a,b], Daniela L. Godoy [a,b], Marcelo G. Armentano [a,b,*], Silvia N. Schiaffino [a,b]

[a] Universidad Nacional del Centro de la Provincia de Buenos Aires, Facultad de Ciencias Exactas, ISISTAN, Campus Universitario, Tandil, 7000, Buenos Aires, Argentina
[b] ISISTAN, CONICET, Campus Universitario, Tandil, 7000, Buenos Aires, Argentina

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ABSTRACT

The use of mobile devices for the study of urban mobility has attracted the attention of the scientific community in the last years. In this work, we address the problem of predicting the user's next destination given the user's mobility profile, i.e. her visit history and the set of places she visited during that time. To predict the next visit, our approach relies on a set of predictors, each of which analyzes different aspects of the user's daily mobility. The output of each predictor consists of a list of places along with the probability of each place being the next destination. The approach combines the output of each predictor into a single one using weights based on the cumulative precision of its previous predictions. Once the user arrives at her destination the proposed approach adapts the set of predictors for future estimations. The predictors in this work were designed so that their training and use have a low computational cost, adapting to the available resources on mobile devices. The experiments carried out over two datasets (Nokia MDC dataset with 52 users and 129,097 visits to 12,869 different places, and a private dataset with 76 users with more than 30 registered days and 47,933 stays) yielded promising results, enhancing performance for 75.8 % of users with an average accuracy of 49.2 % and $F1_{weighted}$ of 42.6 % and an average use of only 48 Megabytes of RAM.

1. Introduction

Understanding and predicting user mobility patterns has become an important research topic due to its relevance in many application domains, including transportation systems, personalized services, and urban planning. In this Introduction, we first describe the motivation behind the short-term mobility prediction problem and the main challenges involved. Then, we outline the contributions of our work in addressing these challenges. Finally, we summarize the structure of the rest of the paper.

1.1. Motivation

In a user mobility analysis scenario, the problem of predicting the next visit of a user has attracted the research community attention and has been addressed with many approaches. In general, mobility models can be divided into *short-term* models which focus on predicting mobility patterns within a relatively short time frame, capturing near future movements, and *long-term* models, which operate on a larger time scale capturing long term trends (e.g. predicting future travel demand) (Garola, Siragusa, Seghezzi, & Mangiaracina, 2024).

The task addressed in this paper focuses on short-term prediction models based on GPS traces (Shams & Haratizadeh, 2018; Solomon, Livne, Katz, Shapira, & Rokach, 2021), which diverges from the typical Point of Interest (POI) prediction commonly applied in the tourism domain (Yu et al., 2025; Zeng et al., 2023). While POI prediction often focuses on discovering new, potentially unexplored locations of interest for tourists, our task involves predicting the next visit of a user within her daily routine. This requires a consideration of the routine nature of the user's mobility, where patterns of repeated visits to familiar locations play a crucial role. In contrast, tourism-related POI prediction emphasizes the exploration of novel destinations rather than the prediction of visits to already known locations.

Different techniques and algorithms have been used in the literature to predict user mobility. Markov models have been widely used (Ashbrook & Starner, 2003; Herder, Siehndel, & Kawase, 2014; Kulkarni, Moro, & Garbinato, 2016), followed by Bayesian networks (Etter, Kafsi, Kazemi, Grossglauser, & Thiran, 2013; Nishino, Nakamura, Yagi, Muto, & Abe, 2010), SVM classifiers (Prabhala, Wang, Deb, La Porta, & Han, 2014; Zolotukhin, Ivannikova, & Hämäläinen, 2013), decision trees (Etter et al., 2013; Gomes, Phua, & Krishnaswamy, 2013) and neural

* Corresponding author.
E-mail address: marcelo.armentano@isistan.unicen.edu.ar (M.G. Armentano).

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networks (Etter et al., 2013; Kulkarni et al., 2016; Zolotukhin et al., 2013), among other techniques (Imai, Tsubouchi, Konishi, & Shimosaka, 2018; Yi, Li, Wang, Zheng, & Sun, 2017; Yu et al., 2017). However, none of the proposed techniques can be applied in all situations and for all users, since the proposed prediction models work well in some cases but not so well in others, depending on the users' mobility patterns. A user's mobility patterns can also change over time, for example if the user moves to another house or changes her workplace. Thus, mobility patterns need some kind of aging or banishing policy that prioritizes the more recent mobility routines. The mobility patterns can even change drastically if the user moves to a different city. In this case, prediction models will have to learn new places and routines from scratch. In addition to these problems, mobile devices highly depend on their battery and have limited memory and processing capacities. Thus, the prediction models should not be computationally expensive in order to be adopted by users.

1.2. Contributions

In this context, in a previous work (Vallejos, Berdún, Armentano, Schiaffino, & Godoy, 2024) we have proposed a lightweight approach to build a user's mobility profile from data collected with mobile devices. The mobility profiles consist of the places visited, the visit history and the travel paths. This previous work aimed to solve some of the challenges and limitations identified in the literature. Particularly, it considers geographic information to identify certain kinds of places, such as open spaces, big places and small places, which are hard to distinguish with existing approaches. We used different sensors and time frequencies to collect data in order to optimize battery consumption and maximize precision. In turn, it executes entirely on mobile devices, avoiding the exposure of sensitive user information and then preserving user privacy. The proposal was evaluated in the context of real usage of the developed prototype applications in two cities of Argentina. The results obtained with our approach outperformed other approaches in the literature, both in precision and recall.

Now, in this work, we present our proposal to predict the user's next visit according to this mobility profile. Our approach is based on a set of prediction models that suggest the next place considering the current place. Each prediction model (corresponding to one region), consists of a set of predictions, and it is trained specifically for that region. In turn, each predictor analyzes different aspects of the user's mobility routines (e.g. day of the week, the time within the day or combinations of these aspects) and it is weighted depending on its cumulative prediction precision. In this way, the approach adapts to each particular user, prioritizing the predictors that best suit her mobility patterns. Predictors have been implemented as probabilistic matrices, with origin as rows and destinations as columns. These prediction models have a low computational cost, in order to be executed on mobile devices without affecting the battery consumption. Our proposal has been experimentally evaluated with two datasets: NOKIA MDC dataset and a private dataset. The Nokia MDC dataset consists of data from 152 users, with 129,097 visits to 12,869 different places. The private dataset consists of records from 76 users with more than 30 registered days and 47,933 stays detected for the participants. Results on these datasets were compared with different baselines models (such as bayesian networks, Markov models, SVM, IBk, Decision Trees, AdaBoostM1, deep learning, and ad hoc probabilistic models), concluding that the proposed approach predicts the user's next destination more accurately.

1.3. Organization

The remainder of this paper is organized as follows. In Section 2 we analyze some related works. In Section 3 we describe our proposed approach to detect the user's next visit. Then, in Section 4 we present the experimental evaluation and the results obtained. In Section 5, we discuss the implications of the results obtained. Finally, in Section 6 we present our conclusions, the limitations of our work and we outline some future research lines.

2. Related work

User mobility prediction has been widely studied, with numerous techniques proposed over the years. In this section, we review the main approaches in the literature, distinguishing between two closely related but distinct problems: next location prediction and next visit prediction. We first present the main techniques and models used for next location prediction based on historical trajectory data. Then, we focus on approaches for next visit prediction, where the objective is to forecast the user's future trips, considering origin, destination, and timing. Throughout the discussion, we highlight the models, features, and contexts considered by different proposals, and conclude with a comparative analysis summarizing the main characteristics of the reviewed approaches.

2.1. Next location prediction

Next location prediction aims to predict the user's possible next visiting location based on the historical mobility sequence data, such as the historical GPS trajectory. Different techniques have been used, such as Markov models (Ashbrook & Starner, 2003; Herder et al., 2014; Kulkarni et al., 2016; Yan, Li, Chan, Shen, & Yu, 2021; Yang, Xu, Xu, Zheng, & Chen, 2014), Bayesian networks (Etter et al., 2013; Nishino et al., 2010; Yanagihara, Namiki, Nawa, Weir, & Oguchi, 2013), SVM classifiers (Prabhala et al., 2014; Zolotukhin et al., 2013), decision trees (Etter et al., 2013; Gomes et al., 2013), and more recently neural networks (Etter et al., 2013; Kulkarni et al., 2016; Zolotukhin et al., 2013), among other approaches (Imai et al., 2018; Yi et al., 2017; Yu et al., 2017). For example, Wu, Chen, Sun, Zheng, and Wang (2017) adopted deep learning techniques (particularly recurrent neural networks, RNN) to model road network constrained trajectories and conducted a comprehensive experimental study based on real datasets to evaluate the performance of their proposal. Comito (2020) proposed NexT, a next-place prediction framework, which exploits LBSNs (Location Based Social Networks) data to forecast the next location of an individual based on the observations of her mobility behavior over some period of time and the recent locations that she has visited. The approach integrates frequent pattern mining and feature-based supervised classification. Yao et al. (2023) proposed a GEMA-BiLSTM (Geographical Embedding and Multi-layer Attention-Bidirectional Long Short-Term Memory) model to predict the next-location in users' mobility. The model combines location and spatio-temporal information to extract the semantics of human mobility. Sun and Kim (2021) proposed deep neural network models that jointly predicted a vehicle's next location and arrival time. They learned sequential patterns in trajectory data using LSTM networks and self-attention mechanisms. For more information, the reader may consult a survey on next location prediction techniques, applications, and challenges (Chekol & Fufa, 2022).

2.2. Next visit prediction

Predicting the next trip aims to predict the next visit information, including the next trip's origin, destination, and time. The mobility data used in the next trip prediction are mobility records. Different techniques and algorithms have been used in the literature to predict the next visit. The technique we found in most works is Markov models (Alvarez-Garcia et al., 2010; Gambs et al., 2012; Song et al., 2006; Zong et al., 2019), followed by Bayesian networks (Dash et al., 2015; Kim et al., 2011; Nguyen et al., 2012; Yanagihara et al., 2013), SVM classifiers (Prabhala & La Porta, 2015), decision trees and more recently neural networks (Liu et al., 2023; Qin et al., 2024; Zhang et al., 2023). For example, Zong et al. (2019) proposed a model for destination prediction by using multi-day GPS data. The habit destination choice is studied in the pre-trip destination prediction model. The during-trip destination prediction model is developed based on the Hidden Markov model. Zhang et al. (2023) proposed DeepTrip, a Deep Learning model for the individual next trip prediction based on a Gated Recurrent Unit (GRU). The DeepTrip model is based on a trip sequence-to-sequence deep learning structure coupled with an attention mechanism. In this sense, Qin et al. (2024) propose DeepAGS, a deep learning-based framework. It is worth noticing that these last models are used to predict next destinations in the context of urban transportation. Liu et al. (2023) propose a graph neural network for long-term preference modeling and LSTM for short-term preference modeling. That study primarily focuses on the users' trajectories throughout the days (historical and current), each consisting of a sequence of visited locations. As regards probabilistic approaches, Kim et al. (2011) proposed AdNext, a visit-pattern-aware mobile advertising system for urban commercial complexes. AdNext predicts the next visit place by learning the sequential visit patterns of commercial complex users in a collective manner. The authors developed a probabilistic prediction model that predicts users' next visit place from their place visit history.

In Yanagihara et al. (2013) the authors proposed a predictor based on a single Bayesian network that considers different attributes of the user's current context. In Dash et al. (2015) the authors propose a combination of eleven Bayesian networks, each of them trained with different attributes. The approaches described in Gao et al. (2012); Nguyen et al. (2012) use ad hoc probabilistic models. Nguyen et al. (2012) compute the probabilities of the next destination based on the user's current location and the time of the day, distinguishing between business days and weekends. When predicting, the authors use empirically determined weights of 0.45 for the current location and 0.55 for the time. Gao et al. (2012) propose a similar probabilistic model, distinguishing among different days in the week. The main difference between these works is that Gao et al. (2012) use a Gaussian distribution for representing the probability that a user travels to a place L in a time T, avoiding the discretization of the time variable. The works described by Gambs et al. (2012), Song et al. (2006) and Yang et al. (2014) use Markov models. In Gambs et al. (2012) the authors use Markov chain of second order. The main problem with this type of predictor is that it is not capable of predicting a sequence of two places the first time. Song et al. (2006) proposed a cascade combination of Markov chains such that, if Markov chains of order N cannot predict a destination, the task relies on Markov chains of order $N-1$, and successively until reaching Markov chains of order 0. The initial chains chosen in this work were second-order Markov chains. Yang et al. (2014) proposed an N-order Markov model with what the authors call an escape probability that falls back to an (N-1)-order Markov model in case the Markov model is unable to make a prediction or calculate the probability that a place L is the next destination. Finally, Prabhala and La Porta (2015) and Zhu et al. (2012) proposed various multi-class classification techniques. In Prabhala and La Porta (2015), a SVM predictor was proposed that considers the place of origin, the time at which the trip begins, the day of the week, whether it is a weekend or not, among other attributes. In Zhu et al. (2012), the authors used different techniques: Naïve Bayes, Bayesian networks, IBk (nearest neighbor), decision trees, and AdaBoostM1. All techniques consider the place of origin, the time at which the trip begins, the day of the week, and the duration of the last visit.

In Table 1, we compare the approaches that predict the next visit proposed in the related works based on the type of model used as a predictor and the attributes used to generate the predictions. Although these techniques have obtained good results, some of them require computational resources that make them not viable for a lightweight approach as ours. For a survey on mobility prediction, the reader can access Ma and Zhang (2022).

3. Proposed approach

This section presents the proposed approach for user destination prediction, detailing its design and the underlying mechanisms that enable

Table 1
Comparison of related work according to the model and attributes used.

Approach	Model	Attributes				
		Day of week	Weekday/Weekend	Time of day	Origin	Others
(Yanagihara et al., 2013)	Bayesian Network	✓		✓		Week of year, trip duration, direction
(Zhu, Sun, & Wang, 2012)	kNN	✓		✓	✓	Duration of the current visit
(Nguyen et al., 2012)	Bayesian Network		✓	✓	✓	Phone usage feature (call log state, ringtone mode, charging state)
(Yang et al., 2014)	Markov				✓	N previous visited locations
(Gao, Tang, & Liu, 2012)	Probabilistic (Hierarchical Pitman-Yor (HPY))	✓		✓	✓	
(Gambs, Killijian, & del Prado Cortez, 2012)	Markov				✓	N previous visited locations
(Dash et al., 2015)	Bayesian Network	✓	✓	✓	✓	Call Detail Records
(Prabhala & La Porta, 2015)	Support Vector Machines	✓		✓	✓	Semantic information, associated with places, demographics information
(Song, Kotz, Jain, & He, 2006)	Markov LZ-based predictors Prediction by Partial Matching (PPM) Sampled Pattern Matching (SPM)			✓	✓	Wi-Fi information
(Zong, Tian, He, Tang, & Lv, 2019)	Markov		✓	✓	✓	Previous GPS points, frequently-visited destinations
(Alvarez-Garcia, Ortega, Gonzalez-Abril, & Velasco, 2010)	Hidden Markov				✓	Previous GPS points
(Kim et al., 2011)	Bayesian Network			✓	✓	Gender Age Current visit duration
(Zhang, Koutsopoulos, & Ma, 2023)	Deep Learning (Gated Recurrent Unit)	✓		✓	✓	Chronological list of trips
(Qin, Zhang, & Ma, 2024)	Deep Learning	✓		✓	✓	Chronological list of trips, frequently-visited stations
(Liu, Chen, Huang, Li, & Min, 2023)	Deep Learning (Graph NN and LSTM)					Chronological list of visit (historical and current), distance between locations
Our approach	Probabilistic matrices	✓	✓	✓	✓	

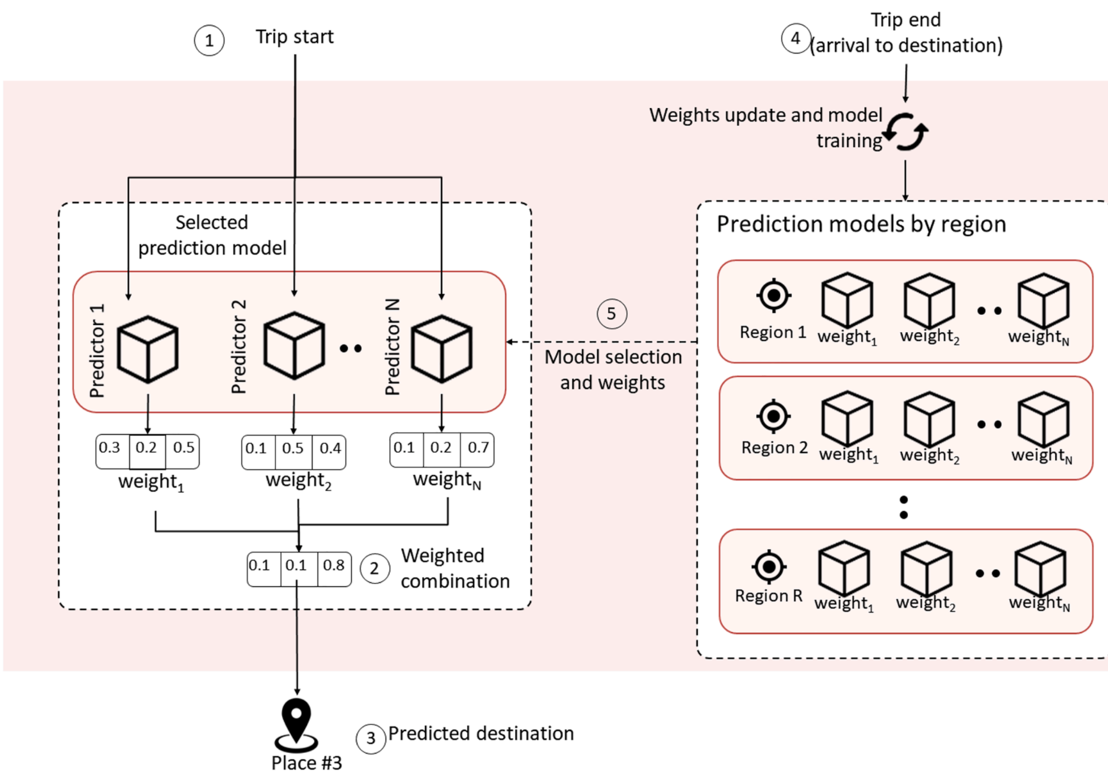

Fig. 1. Overview of the prediction approach.

adaptation to changes in user routines while maintaining low computational complexity. The section is organized as follows. First, we introduce the core component of the approach: a set of lightweight predictors that model different aspects of user mobility patterns. Then, we describe how the predictions generated by these models are combined using an adaptive weighting mechanism based on their historical performance. Finally, we explain the aging strategy implemented to prioritize recent mobility patterns over older ones. Fig. 1 provides an overview of the proposed method.

3.1. Overview

An overview of our approach is depicted in Fig. 1. Each time a user starts a new trip (marked as 1 in Fig. 1), the approach predicts the next destination considering the user's recorded trips. To achieve this task, it relies on different predictors, i.e. functions that map different aspects of the users' context (e.g. origin, day of the week, time) to the potential place to be visited. In this work, we have defined nine predictors, each of which considers a different aspect of the user's mobility routine. In turn, predictors have an aging mechanism so that new mobility patterns have more weight than older ones. In this way, predictors can adapt to changes in the user's routines. Also, predictors are designed so that their training and usage have a low computational cost to adapt to mobile device resources.

Each prediction consists of a list of places with their associated probabilities of being the next destination. The approach combines the outputs of each predictor in a combined one (marked as 2 in Fig. 1) using a weighting mechanism. The weight of each predictor depends on the cumulative prediction precision. As the precision might vary depending on the day of the week (e.g. a predictor might work well for working days but wrong for weekends), we keep different precision values for the same predictor depending on the day of the week and the time of the day. The next destination is predicted as the one with the highest combined probability (marked as 3 in Fig. 1). This destination can be used by various services or applications to assist the user in her daily routine. When the user arrives at the destination (marked as 4 in Fig. 1) it is compared to the destination predicted by each predictor and the cumulative precision values are updated. Predictors are organized according to the region the user is in into prediction models by region. Thus, when the user starts a new trip the approach selects the prediction model according to the region the user is in (marked as 5 in Fig. 1).

Table 2
Organization of the predictors according to the considered aspects of the user mobility.

Aspect of the user mobility	Predictors	# matrices
Transition	Day of the week	7
	Weekend and Working days	2
	Full week	1
Temporal	Day of the week	7
	Weekend and Working days	2
	Full week	1
Stay	Day of the week	7
	Weekend and Working days	2
	Full week	1

3.2. Predictors

Predictors were implemented as probabilistic matrices, so that their training and use are efficient enough to run on the mobile devices of the users. Nine predictors were implemented. These predictors are organized into three main groups according to three different aspects of the user mobility, as shown in Table 2.

3.2.1. Transition matrix

Transition matrices analyze user mobility spatially by predicting the user's next destination based on their current location. The operation of transition matrices is similar to that of Markov chains (Wilinski, 2019). Specifically, the transition matrix stores counters for the number of transitions between each pair of locations and the total outgoing transitions from each location. Sub-figure (a) of Fig. 2 shows an example matrix

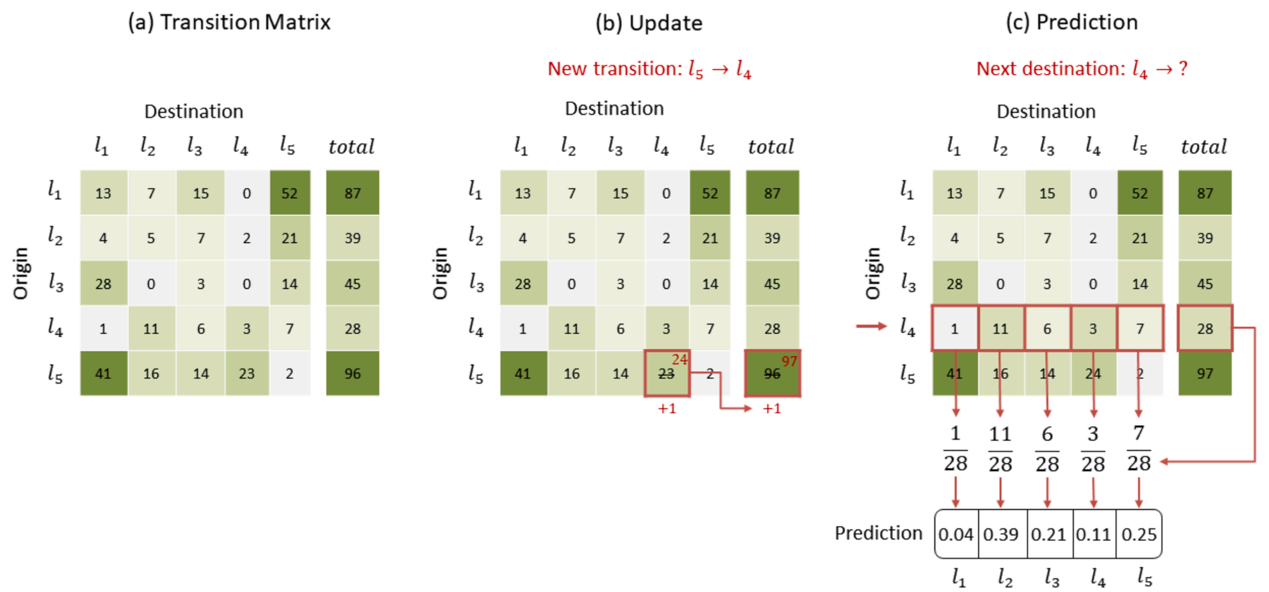

Fig. 2. Example of a transition matrix update.

built from transitions between five locations. Each time the user moves from one place to another, our approach updates the matrix by simply increasing the value of the corresponding cells, as shown in sub-figure (b). Additionally, when a trip begins, our approach makes a prediction by dividing the values in the row corresponding to the origin location of the transition by the total outgoing transitions from that origin. Sub-figure (c) demonstrates how the transition matrix predicts the next destination upon detecting that the user initiates a trip from location l_4.

One of the disadvantages of transition matrices is that they only consider the spatial aspect (where the user is) when predicting and do not account for the temporal aspect (for example, what day it is). To consider the temporal aspect, time can be divided into bins, creating a transition matrix for each bin. For example, a matrix could be created for each day of the week. Each matrix is updated only based on trips made during that specific day and is used to predict when the user initiates a trip on that particular day of the week. A predictor of this nature allows for capturing day-specific routines. However, the more specific the predictor, the longer it takes to learn, since it will have fewer training examples.

Considering this, three predictors based on transition matrices were defined:

- The first predictor uses seven transition matrices, one for each day of the week. This predictor is the most specific and, as previously mentioned, takes the longest to learn.
- The second predictor uses two transition matrices: one for weekdays and another for weekends. This is because the places the user visits during the week often differ from those visited on weekends.
- The third predictor uses a single transition matrix. This is the simplest predictor and learns the fastest, as it trains its single matrix with all transitions made by the user.

3.2.2. Temporal matrix

Temporal matrices analyze user mobility temporally, as they predict the user's next destination based on the time the trip starts. The temporal matrix stores a count of the number of transitions to each location according to the time of day. To achieve this, it is necessary to discretize the 24 hours of the day into bins. Specifically, in this work, it was empirically found that 36 bins of 40 minutes were sufficient. Sub-figure (a) of Fig. 3 shows an example matrix built from transitions between five locations. Each row represents one of the 36 temporal bins, while each column represents a different location. Each time the user moves from one place to another, this approach adds information from the new transition to the matrix by simply updating the values in the corresponding cells. Sub-figure (b) shows how the matrix is updated after a trip beginning at 11:30 pm (which, when discretized, falls into bin 35) with destination l_4.

At the start of a trip, the approach generates a prediction by dividing the values in the row corresponding to the bin of the trip's start time by the total number of transitions that occurred in that bin. Since time is a continuous variable, dividing it into bins introduces a discretization error (when two very close values fall into different bins). For this reason, the temporal matrix considers not only the bin corresponding to the start time of the transition but also its two neighboring bins. Sub-figure (c) shows how the temporal matrix predicts a next destination when it detects that the user starts a trip at 1:00 a.m. (which, when discretized, falls into bin 2).

Similar to transition matrices, the temporal matrix only considers the time of day but it does not account for, for example, the day of the week. Knowing the day is crucial when looking for temporal mobility patterns. For instance, if the user starts a trip at 9 a.m. on a weekday, it is likely that she is going to work. However, if this happens on a weekend, she might be going to church or another type of activity. There are also specific routines for each day. For example, the user might go to the gym on Tuesdays and Thursdays at 6 p.m., while on other days at the same time, she might attend an Spanish class. Considering this, three predictors based on temporal matrices were defined:

- The first predictor uses seven temporal matrices, one for each day of the week.
- The second predictor uses two temporal matrices: one for weekdays and another for weekends.
- The third predictor uses a single temporal matrix.

3.2.3. Stay matrix

Stay matrices analyze user mobility temporally, but differ from the previous matrices as they are not trained based on the trips the user makes from one place to another, but rather on the stays. The stay matrix does not evaluate where the user typically travels at a given time, but rather where she is likely to be at that moment. Suppose that the user typically works from 9:00 a.m. to 5:00 p.m., but one day she takes the morning off to run an errand. At 10:30 a.m., the user finishes the errand and heads to work. The issue is that, in this situation, temporal matrices cannot predict the user's destination because the user does not typically

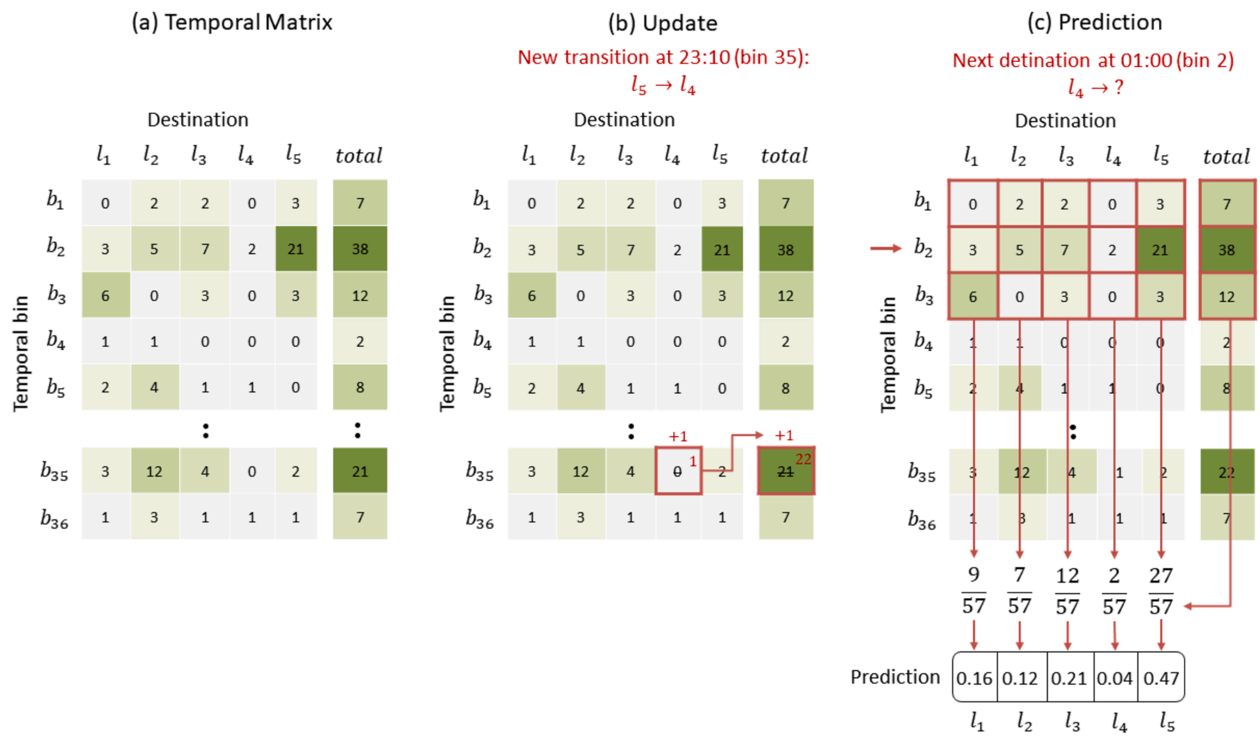

Fig. 3. Example of a temporal matrix update.

travel to work at this time. By considering where the user is likely to be, stay matrices are able to identify that the user is typically at work at that time and predict that the user is heading to work.

The stay matrix stores the number of stays in each location according to the time of day. Similar to temporal matrices, the 24 hours of the day were divided into 36 bins of 40 minutes each. Sub-figure (a) of Fig. 4 shows an example matrix built from stays in five locations. Each row represents one of the 36 temporal bins, while each column represents a different location. Every time the user finishes a stay, the values in the corresponding cells are updated. Sub-figure (b) shows how the matrix is updated after a stay in l_4 between 11:30 p.m. (bin 36) and 1:00 a.m. (bin 2). Note that bin 17 is also updated, since the user was in location l_4 during that time.

At the start of a trip, the approach generates a prediction by dividing the values in the row corresponding to the bin of the trip's start time by the total number of transitions that occurred in that bin. As mentioned earlier, dividing the time into bins introduces a discretization error. For this reason, the temporal matrix considers not only the bin corresponding to the start time of the transition, but also the next bin. Sub-figure (c) shows how the temporal matrix predicts a next destination when it detects that the user starts a trip at 1:00 a.m. (bin 2). Note that the previous bin (bin 1) is not considered in the calculations, while the next bin (bin 3) is considered. This is because the user started their trip in bin 2, and it is assumed that the user will arrive at their destination in the near future. For this reason, the stay matrix considers the next bin but not the previous one.

Similar to the transition matrices and temporal matrices, the stay matrix does not consider the day of the week by itself, which is crucial for identifying patterns in user mobility. For this reason, and similarly to the other two types of matrices, three predictors based on stay matrices were defined:

- A first predictor that uses seven stay matrices: one for each day of the week.
- A second predictor that uses two stay matrices: one for weekdays and another for weekends.
- A third predictor that uses a single stay matrix.

3.3. Weighting and combination

The approach combines the destination predictions made by each of the predictors described in Section 3.2. At this stage, the approach weights the predictions based on the accumulated accuracy of each predictor. Since some predictors learn and predict the mobility of certain users better than others, weighting the predictors according to the accuracy of their previous predictions allows the approach to prioritize those predictors that best predict the user.

Fig. 5 shows the prediction combination flow. The probabilities of each prediction are multiplied by the accumulated precision (Precision_{acc}) of its respective predictor (marked with 1 in Fig. 5). The resulting probabilities of each prediction are summed into a single prediction (marked with 2 in Fig. 5). This prediction is normalized so that all probabilities are between 0 and 1, and their total sums to 1. To achieve this, each probability is divided by the total sum of all probabilities (which results in 1.31 in the example). The resulting probabilities constitute the final prediction of the approach (marked with 3 in Fig. 5). The location with the highest probability of being the user's next destination is the predicted destination. In the example shown in Fig. 5, the approach predicts the location l_5 with a probability of 0.37.

The accuracy values used by the approach are updated each time the user reaches her destination. In the example shown in Fig. 5, the user's actual destination was location l_5. At this point, the approach compares the actual destination the user arrived at (l_5) with the destination predicted by each individual predictor. In this example, the predictor by origin and time predicted location l_2 as the most likely one, while the other two predictors predicted location l_5. Thus, the first predictor accumulates an error, while the other two predictors each accumulate a correct prediction. The updated count of correct predictions and errors is then used to calculate the new accuracy of each predictor.

Since user mobility varies throughout the week and weekends, or even between morning and evening on the same day, the accuracy of each predictor can fluctuate over time. For instance, during weekdays, users often follow structured schedules for work, gym, etc.

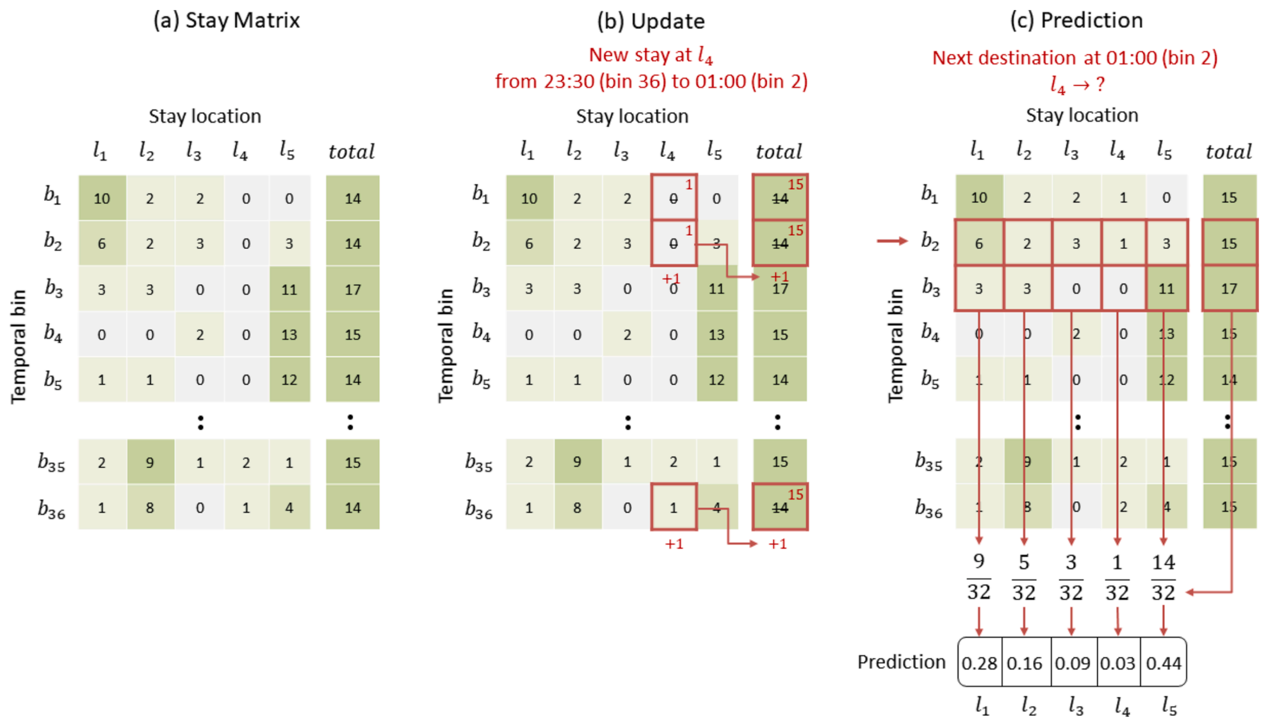

Fig. 4. Example of a stay matrix update.

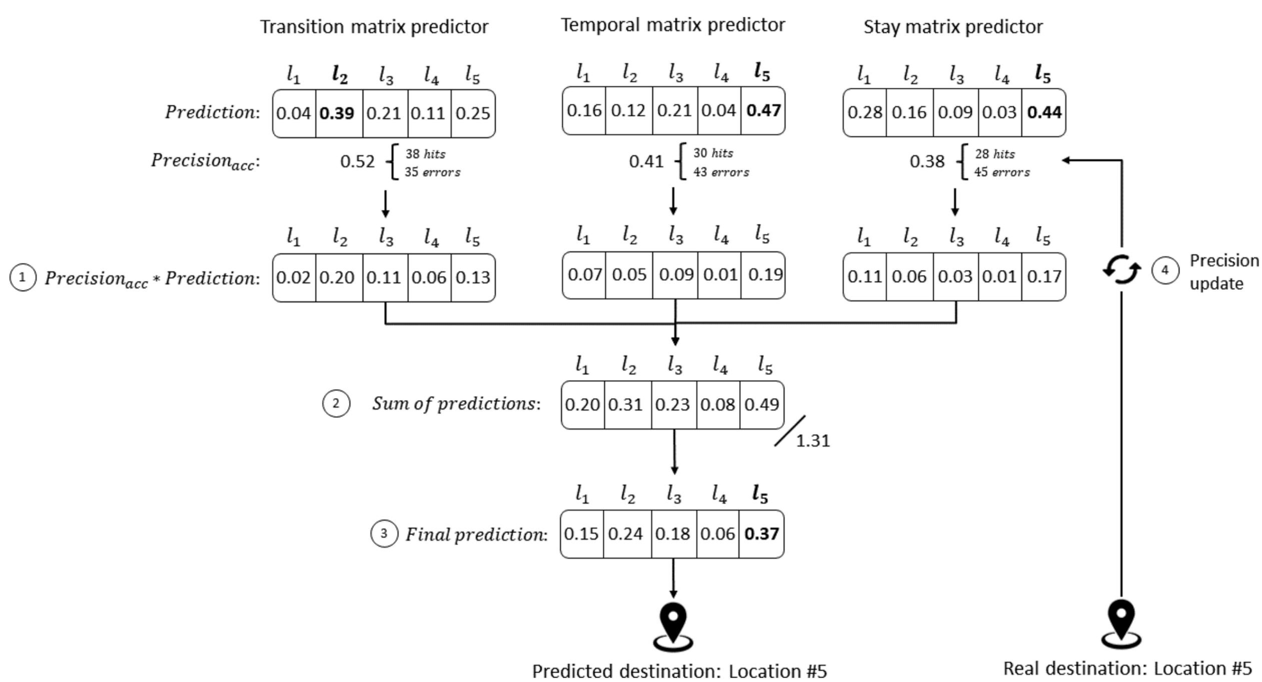

Fig. 5. Combination of predictions.

Thus, a temporal matrix might accurately predict user mobility during weekdays. However, on weekends, users typically have less structured schedules, so in this situation, a transition matrix might be better suited than a temporal matrix. Considering this, the approach uses four accuracy weights for each predictor, distinguishing between weekdays and weekends, and between morning (00:00–12:00) and afternoon/evening (12:00–24:00). This way, the approach can adapt by prioritizing predictors that best fit each particular user and each time of the week.

3.4. Aging mechanism

To allow predictors to adapt to users' routine changes, a mechanism was employed that weighs new transitions over old ones. For this, the proposed approach allows setting an aging value ($\alpha > 0$) with which the values used to update the predictors' matrices are modified day by day. Initially, the update value is set to 1, so on the first day, all transitions or stays with which predictors update their matrices are assigned the value 1. For example, if the user travels from $l_2 \rightarrow l_4$, the transition matrix updates the corresponding cell by increasing its value by 1. As the days go by, the predictors adjust the update value according to Eq. (1)

$$update_{value} = update_{value} * (1 + \alpha)) \quad (1)$$

This way, each new day of user mobility is weighted α times more than the previous day. On day N, the predictors will update their matrices with the update value $1 * (1 + \alpha)^N$. This gradually prioritizes the most recent transitions, as they affect the matrix more than transitions from previous days. This aging mechanism not only has low computational cost but is also incremental, avoiding the need for retraining prediction models as with other mechanisms. In this particular work, the value of $\alpha = 0.035$ was empirically set. This means that on the first day, the update value is 1, on the second day, the second transition uses the value 1.035, on the third day, it is 1.071, and so on. With the value α established in this work, the mobility data recorded each month is weighted 2.62 times more than the mobility data recorded in the previous month, allowing the approach to forget old routines and learn new ones (assuming that routines typically exhibit monthly variations). Smaller values would cause the approach to take several months to learn new routines, while larger values would hinder the learning process.

4. Materials and methods

In order to validate our proposed approach, we conducted a series of experiments using two real-world datasets. This section details the datasets employed, the baseline methods selected for comparison, the metrics used to assess the predictive performance, and the experimental results obtained. We also present an ablation study to analyze the contribution of each component of our approach.

4.1. Dataset

We have used different datasets to evaluate our proposal, the NOKIA MDC dataset and a private dataset used in our previous work (Vallejos et al., 2024). The dataset known as Nokia MDC consists of 152 users, 129,097 visits and 12,869 different places. Currently, Nokia MDC dataset is probably one of the most complete public datasets in the urban mobility domain. Visits and places are not uniformly distributed among users. Fig. 6 shows the frequency of users with respect to places and visits. In Fig. 6(a) and (b), we can observe that most users have between 500 and 1000 visits and between 50 and 100 detected places. However, there exist users with more than 1000 visits, while others have less than 500 visits. Fig. 6(c) and (d) show the number of visits and detected places for each user with respect to the number of days in which their movements were registered. As expected, there is a tendency to detect more places and visits as the users participated more days in the experiment. However this is not strict since some users have the same number of visits than others with the double of registered days. This is strongly related to the users´ routines and personality, since some users are more active and predictable than others. We can also observe that the rate at which new places are discovered decreases with time. For example, if we split the visits history in two, during the first part 12.47 % of visits are to new places ("discovering visits") while in the second part visits to new places correspond to 6.68 % of the total number of visits. In this work, the Nokia MDC dataset was split in two, according to the duration of stays sequences: one sequence of 107,293 stays lasting more than 20 minutes (Nokia-20), and another sequence of 123,721 stays lasting more than 10 minutes (Nokia-10). Moreover, we eliminate 10 users with less than 30 registered days or less than 30 trips.

Table 3
Dataset statistics.

Datasets used	# Users	# Stays	# Days	# Stays/day
Nokia MDC - 20 min	142	107,293	39,737	2.7
Nokia MDC - 10 min	142	123,721	39,737	3.1
Private dataset	76	47,315	7908	5.9

The other dataset used is a private dataset collected with the prototype presented in Vallejos et al. (2024). It consists of records from 76 users with more than 30 registered days and more than 30 trips. The dataset contains 47,933 stays detected for the participants.

Table 3 shows a summary of the three datasets used for the experiments. With these datasets, we test our approach on the nine predictors detailed in Section 3, with respect to different baselines presented in Section 4.2. We also perform an ablation study to test the benefits of each component of the approach.

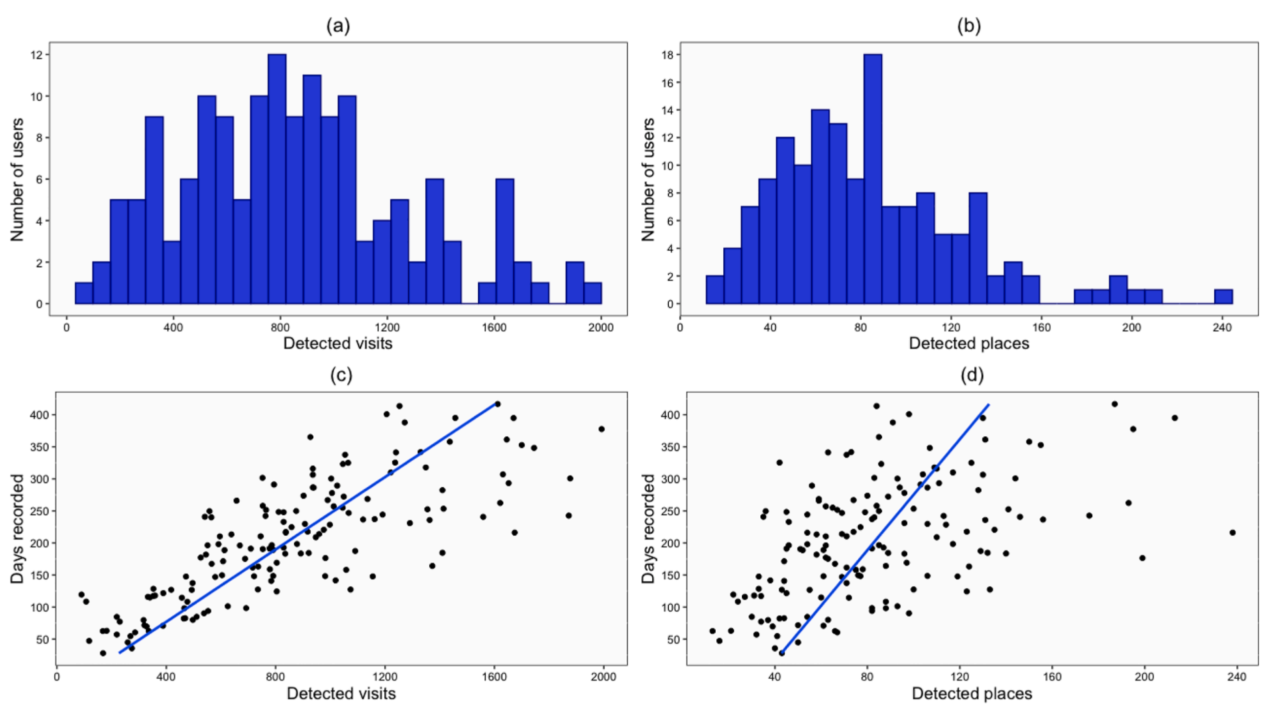

Fig. 6. Distribution of visits and places for the users in the Nokia MDC dataset.

4.2. Baselines

We decided to compare our approach with respect to different predictors used in the literature. Some of the approaches are proposed in Yanagihara et al. (2013), Zhu et al. (2012), Nguyen et al. (2012), Yang et al. (2014), Gao et al. (2012), Gambs et al. (2012), Dash et al. (2015), Song et al. (2006) and Prabhala and La Porta (2015), and they consider the prediction model proposed by Laurila et al. (2013) as a baseline. This baseline simply consists in predicting the users' most visited place as the next destination. If the user is already at the most visited place, then the second most visited one is predicted. Moreover, we compare the results with the deep learning-based approach proposed by Liu et al. (2023), which in turn outperforms the approaches proposed by Feng et al. (2018); Liu, Wu, Wang, and Tan (2016); Sun et al. (2020).

4.3. Evaluation metrics

In the problem of predicting the next destination, the task is considered as a multi-class classification problem where each potential destination is treated as a distinct class. To evaluate the model's performance, common metrics include Accuracy, Precision, Recall, and the F1-score (Manning, Raghavan, & Schütze, 2008). Accuracy is calculated as a single value that aggregates performance over all classes, while Precision, Recall, and F1-score are computed in macro, and weighted forms.

Accuracy measures the overall proportion of correctly classified instances, as defined by eq. 2, where TP_i represents the number of true positives for class i and N is the total number of samples.

$$\text{Accuracy} = \frac{\sum_{i=1}^{C} TP_i}{N} \tag{2}$$

While accuracy provides a general assessment of performance, it can be misleading in imbalanced datasets, as a high accuracy might be achieved by merely predicting the majority classes correctly.

For Precision, the macro-average precision is calculated by taking the simple average of the precision for each class (Eq. (3)).

$$P_{\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} \frac{TP_i}{TP_i + FP_i}. \tag{3}$$

The weighted precision adjusts for class imbalance by weighting each class by its proportion w_i, where $w_i = \frac{N_i}{N}$, with N_i being the number of examples in class i and N the total number of examples, (Eq. (4)).

$$P_{\text{weighted}} = \sum_{i=1}^{C} w_i \cdot \frac{TP_i}{TP_i + FP_i}. \tag{4}$$

Similarly, the recall metrics are given by Eqs. (5), and (6).

$$R_{\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} \frac{TP_i}{TP_i + FN_i} \tag{5}$$

$$R_{\text{weighted}} = \sum_{i=1}^{C} w_i \cdot \frac{TP_i}{TP_i + FN_i} \tag{6}$$

The F1-scores, which represent the harmonic mean of precision and recall, are given by Eqs. (7), and (8).

$$F1_{\text{macro}} = \frac{1}{C}\sum_{i=1}^{C} \frac{2 \cdot Precision_i \cdot Recall_i}{Precision_i + Recall_i} \tag{7}$$

$$F1_{\text{weighted}} = \sum_{i=1}^{C} w_i \cdot \frac{2 \cdot Precision_i \cdot Recall_i}{Precision_i + Recall_i} \tag{8}$$

These metrics provide a comprehensive evaluation of the classifier's performance across all classes, especially crucial in imbalanced multi-class scenarios.

4.4. Results

In this section we first study the impact of the different components of the approach on the performance (Section 4.4.1), and then we compare our proposal against different baselines and approaches found in the literature (Section 4.4.2).

4.4.1. Ablation study

Table 4 summarizes the results obtained by the proposed approach and its reduced versions. The first column indicates the predictors used, while the remaining columns show the metrics computed for all users in the three datasets detailed in Table 3. The table presents the results of the complete approach compared to its simplified versions: without aging, without regions, and using only stay matrices, temporal matrices, and transition matrices.

The ablation study highlights the contribution of different components to the overall performance of the approach. The full version (first row), achieves the best results across all metrics, with an accuracy of 49.2 % and an $f1_{weighted}$ score of 42.6 %. This suggests that the combination of all predictors and features provides the most effective model.

When removing regional information (Approach without regions), there is a slight decline in performance, with accuracy dropping to 49.1 % and $f1_{weighted}$ decreasing to 41.8 %. This indicates that regional features contribute to the model's effectiveness but are not critical, as the overall performance remains relatively stable. This can be explained by the fact that very few users register movements among cities.

The individual predictors (Temporal Matrix Predictor Only, Transition Matrix Predictor Only, and Stay Matrix Predictor Only) perform worse compared to the full approach, showing lower accuracy (ranging from 39.5 % to 43.5 %) and weaker $f1_{weighted}$ scores. Among them, the Stay Matrix Predictor Only performs slightly better, suggesting that stay-based transitions might provide more informative patterns than the others when used in isolation.

Interestingly, removing the aging factor (Without aging) results in a noticeable performance drop compared to the full approach, with accuracy decreasing from 49.2 % to 47.9 %. This suggests that accounting for aging effects in the model plays a key role in improving predictive performance.

In conclusion, the results confirm that the full combination of predictors and features yields the best performance, with regional information and aging effects contributing to improvements. The study also shows that individual predictors alone are not as effective, highlighting the importance of integrating multiple predictive components.

4.4.2. Comparison with related works

Table 5 summarizes the results obtained by the proposed approach and other approaches proposed in the literature by Yanagihara et al. (2013), Zhu et al. (2012), Nguyen et al. (2012), Dash et al. (2015), Yang

Table 4
Results of the approach and ablation study.

Approach	Accuracy	P_{macro}	$P_{weighted}$	R_{macro}	$R_{weighted}$	$F1_{macro}$	$F1_{weighted}$
Approach (full)	0.492	0.052	0.394	0.048	0.492	0.046	0.426
Approach Without Aging	0.479	0.043	0.369	0.041	0.479	0.037	0.398
Approach Without Regions	0.491	0.050	0.385	0.044	0.491	0.042	0.418
Stay Matrix Predictor Only	0.435	0.041	0.367	0.039	0.435	0.036	0.383
Temporal Matrix Predictor Only	0.426	0.039	0.349	0.038	0.426	0.037	0.377
Transition Matrix Predictor Only	0.395	0.039	0.355	0.036	0.395	0.035	0.366

Table 5
Comparison with related works for users in all datasets (360 users).

Authors	Predictor	Accuracy	P_{macro}	$P_{weighted}$	R_{macro}	$R_{weighted}$	$F1_{macro}$	$F1_{weighted}$
Approach	Probabilistic matrices	**0.492**	**0.053**	0.395	**0.048**	**0.492**	**0.046**	**0.426**
Dash et al. (2015)	Bayesian network	0.450	0.028	0.317	0.030	0.450	0.025	0.354
Gambs et al. (2012)	Markov	0.311	0.033	0.343	0.025	0.311	0.026	0.317
Gao et al. (2012)	Probabilistic	0.403	0.042	0.374	0.039	0.403	0.037	0.377
Liu et al. (2023)	Deep Learning	0.173	0.023	0.104	0.039	0.173	0.026	0.120
Laurila et al. (2013)	Most frequent	0.405	0.016	0.269	0.025	0.405	0.019	0.310
Nguyen et al. (2012)	Probabilistic	0.450	0.037	0.344	0.036	0.450	0.033	0.375
Prabhala and La Porta (2015)	SVM	0.397	0.013	0.236	0.021	0.397	0.014	0.268
Song et al. (2006)	Markov	0.434	0.035	0.331	0.035	0.434	0.032	0.365
Yanagihara et al. (2013)	Bayesian network	0.467	0.034	0.351	0.036	0.467	0.032	0.388
Yang et al. (2014)	Markov	0.435	0.035	0.330	0.035	0.435	0.032	0.364
Zhu et al. (2012)	AdaBoostM1	0.411	0.016	0.259	0.025	0.411	0.018	0.305
	Decision Tree	0.455	0.038	0.349	0.037	0.455	0.033	0.377
	Bayesian network	0.467	0.034	0.351	0.036	0.467	0.032	0.388
	IBK	0.379	0.048	**0.398**	**0.048**	0.379	0.045	0.383
	Naive Bayes	0.385	0.032	0.350	0.035	0.385	0.031	0.354
	SVM	0.387	0.019	0.260	0.023	0.387	0.018	0.291

et al. (2014), Song et al. (2006), Gao et al. (2012), Prabhala and La Porta (2015), Gambs et al. (2012), Liu et al. (2023) and the baseline proposed by Laurila et al. (2013).

The results demonstrate that our proposed approach outperforms all state-of-the-art methods across most evaluation metrics. With an accuracy of 49.2 % and an $f1_{weighted}$ score of 42.6 %, our method shows a clear improvement over the competing approaches. Notably, the closest competitor, (Yanagihara et al., 2013), achieves an accuracy of 46.7 % and an $f1_{weighted}$ score of 38.8 %, which are both significantly lower than our approach's performance.

Due to the non-normal distribution of some samples (verified using Shapiro-Wilk test), we employed the Wilcoxon signed-rank test to check the statistical significance of the reported results. The results of the Wilcoxon signed-rank test showed that all observed differences between our approach and the baselines are statistically significant ($p < 0.05$), except for R_{macro} and $F1_{macro}$ when compared to the IBK approach proposed by Zhu et al. (2012). For these two metrics, the differences are not statistically significant, suggesting that while our model performs better overall, IBK achieves comparable results in terms of recall and F1 in their macro version.

The superiority of our approach is evident when compared to classical baselines such as most-frequent-baseline (accuracy: 40.5 %, $F1_{weighted}$: 31.0 %) and more recent baselines, such as Liu et al. (2023) (accuracy: 17.8 %, $F1_{weighted}$: 12.4 %). That approach does not achieve a good performance, although it performs better than the results reported in the original article for other datasets (Liu et al., 2023). Additionally, other methods, including Dash et al. (2015) (accuracy: 45.0 %, $F1_{weighted}$: 35.4 %) and Zhu et al. (2012) decision trees (accuracy: 45.5 %, $F1_{weighted}$: 37.7 %), also show lower performance, further validating the effectiveness of our approach.

In conclusion, our method significantly outperforms all state-of-the-art predictors across most evaluation metrics, with the exception of R_{macro} and $F1_{macro}$ when compared to IBK proposed by Zhu et al. (2012), where the differences are not statistically significant. These results highlight the robustness of our approach and its potential impact in improving predictive performance.

Table 6 shows the results for Accuracy (Acc.), and F-measure macro ($F1_m$) and F-measure weighted ($F1_w$) for each dataset separately. We can observe that our approach achieves the highest values, consistently with the results shown in Table 5. This shows that the combination of different predictors that analyze different aspects of the user's daily mobility achieves better results than a single predictor that considers all aspects simultaneously. With respect to the other predictors, Bayesian networks and decision trees generally achieve better results than other prediction models. The Bayesian network used by Yanagihara et al. (2013) and that proposed by Zhu et al. (2012) obtained almost the same results, since they use almost the same features during training: Zhu et al. approach adds the duration of the user's current stay as an extra feature, but the results show that this characteristic does not influence the prediction. The probabilistic predictor proposed in Nguyen et al.

Table 6
Comparison with related works for each dataset.

Authors	Predictor	Private			Nokia-10			Nokia-20		
		Acc	$F1_m$	$F1_w$	Accuracy	$F1_m$	$F1_w$	Acc	$F1_m$	$F1_w$
Approach	Probabilistic matrices	**0.389**	**0.035**	**0.302**	**0.499**	0.046	**0.434**	**0.540**	**0.052**	**0.484**
Dash et al. (2015)	Bayesian network	0.348	0.021	0.245	0.458	0.025	0.362	0.497	0.028	0.404
Gambs et al. (2012)	Markov	0.165	0.019	0.188	0.331	0.027	0.331	0.369	0.029	0.370
Gao et al. (2012)	Probabilistic	0.271	0.030	0.259	0.422	0.039	0.391	0.456	0.040	0.426
Liu et al. (2023)	Deep Learning	0.150	0.029	0.105	0.178	0.022	0.119	0.181	0.028	0.130
Laurila et al. (2013)	Most frequent	0.339	0.018	0.231	0.404	0.017	0.307	0.440	0.021	0.356
Nguyen et al. (2012)	Probabilistic	0.336	0.030	0.265	0.461	0.033	0.383	0.499	0.035	0.426
Prabhala and La Porta (2015)	SVM	0.311	0.012	0.168	0.401	0.013	0.274	0.440	0.016	0.317
Song et al. (2006)	Markov	0.327	0.028	0.259	0.443	0.033	0.372	0.482	0.034	0.414
Yanagihara et al. (2013)	Bayesian network	0.356	0.026	0.269	0.476	0.031	0.398	0.518	0.035	0.442
Yang et al. (2014)	Markov	0.328	0.028	0.258	0.443	0.033	0.372	0.483	0.034	0.414
Zhu et al. (2012)	AdaBoostM1	0.332	0.017	0.218	0.413	0.016	0.305	0.452	0.020	0.352
	Decision Tree	0.355	0.027	0.260	0.462	0.034	0.388	0.502	0.036	0.428
	Bayesian network	0.356	0.026	0.269	0.476	0.031	0.398	0.518	0.035	0.442
	IBK	0.240	0.032	0.249	0.393	**0.047**	0.396	0.439	0.051	0.441
	Naive Bayes	0.235	0.024	0.220	0.398	0.031	0.366	0.453	0.034	0.414
	SVM	0.316	0.016	0.200	0.386	0.017	0.293	0.426	0.021	0.337

Table 7
Global results for the different approaches.

Authors	Predictor	Memory (MB)	Running time (Seconds)	Optimal users	$F1_w$
Approach	Probabilistic matrices	48	< 0.1	273 (3)	0.426
Dash et al. (2015)	Bayesian network	90	0.5	4 (0)	0.354
Gambs et al. (2012)	Markov	4.1	< 0.1	0 (0)	0.317
Gao et al. (2012)	Probabilistic	24	< 0.1	9 (0)	0.377
Liu et al. (2023)	Deep Learning	44	108	3 (0)	0.120
Laurila et al. (2013)	Most frequent	6	< 0.1	6 (0)	0.310
Nguyen et al. (2012)	Probabilistic	8	< 0.1	4 (0)	0.375
Prabhala and La Porta (2015)	SVM	55	1.6	2 (2)	0.268
Song et al. (2006)	Markov	7	< 0.1	6 (3)	0.365
Yanagihara et al. (2013)	Bayesian network	36	< 0.1	11 (11)	0.388
Yang et al. (2014)	Markov	70	0.5	5 (3)	0.364
Zhu et al. (2012)	AdaBoostM1	56	0.2	1 (0)	0.305
	Decision Tree	156	0.4	18 (3)	0.377
	Bayesian network	37	< 0.1	11 (11)	0.388
	IBK	24	< 0.1	22 (2)	0.383
	Naive Bayes	46	< 0.1	2 (0)	0.354
	SVM	46	2.7	3 (1)	0.291

(2012) obtains competitive results by weighting the destination prediction based on the user's origin location with the destination prediction based on the time of day at which the user begins to move. However, as mentioned previously, this combination of two prediction models has a static weighting system and it does not adapt to each individual user.

The IBK approach achieved higher $f1_{macro}$ values for the Nokia 10 dataset. However, the differences in this metric were not statistically significant according to the Wilcoxon signed-rank test. The same trend was observed for the other two datasets, where our approach obtained better $F1_{macro}$ scores with respect to IBK, but the differences were not statistically significant. This suggests that the improvements for $F1_{macro}$ cannot be considered conclusive across the datasets. All other differences are statistically significant with $p < 0.05$.

Following these results, an attempt was made to add some of the predictors in the literature to the set of predictors of the proposed approach. However, the results did not show a significant gain in accuracy. This can be considered as an indicator that the indiscriminate combination of predictors does not necessarily imply an improvement in prediction. When adding a predictor to the approach, it should be considered that the new predictor addresses the destination prediction problem differently from the predictors already considered in the approach, in search of mobility patterns that the other predictors cannot detect. An example of indiscriminate combination can be seen in Dash et al. (2015), where the authors combined 11 Bayesian networks without achieving significant improvements in the results. Markov-based models (Song et al., 2006; Yang et al., 2014) achieved competitive results, but always below the proposed approach, Bayesian network-based models (Dash et al., 2015; Yanagihara et al., 2013; Zhu et al., 2012) or combinations of other models (Dash et al., 2015; Nguyen et al., 2012). At this point, the model proposed by Gambs et al. (2012) ranked below the other predictors in terms of F-measure, but not due to low accuracy in predicting, but rather due to the number of times it was unable to predict a destination. Finally, some multi-class classification algorithms, such as SVM, IBK, and AdaBoostM1, seemed to not adapt well to the urban mobility domain. On some occasions, these predictors were even surpassed by the baseline proposed in Laurila et al. (2013).

Table 7 shows the average memory consumption of each predictor for all users, along with the time consumption [1]. The $F1_w$ column is added as a reference to the model performance. As expected, decision trees tend to consume more memory than other techniques, while deep learning models tend to consume more time, specially due to the training time. Consequently, these techniques may be impractical, especially if a large amount of user data is accumulated over the long term, making model training increasingly burdensome. Regarding the other predictors, most of them have viable memory consumption for execution on mobile devices. As expected, the predictors with the least requirements are the baseline and those models based on Markov. As for the proposed approach, its execution required an average of 48 Megabytes of RAM (Random Access Memory). This is a reasonable memory requirement considering that the approach internally consists of nine matrix-based predictors.

Additionally, the fifth column of the table indicates for how many of the users each predictor outperformed the others (i.e., obtained a higher $F1_w$ value). For example, the approach presented in Song et al. (2006), outperformed all other predictors for 6 of the users. In some cases, it happens that for a user there is not a single superior predictor, but rather two or more predictors that obtained the same result. These cases are specified between parentheses in Table 7. In the same example of the approach by Song et al. (2006), it was the best predictor along with another predictor in 3 of the 6 users. As it can be seen, the proposed approach was superior in 273 of the 360 users (116 from Nokia-20, 111 from Nokia-10, and 46 from the Private Dataset). This indicates that there are still a few users to whom the approach does not adapt as well as other predictors do. It is worth noticing that, regardless of the overall results obtained, almost all predictors outperform the others in some small group of users. Even the baseline, despite its simplicity, is superior for 6 of the users. This type of analysis can be useful to identify which type of predictors is convenient to combine.

5. Discussion

The experimental evaluation of this work demonstrated the capacity of the proposed approach for studying the urban mobility of citizens and its superiority over other approaches proposed in the literature in predicting the next location that the user might visit.

We have shown that, by combining different predictors with a weighting mechanism that adapts to each user in particular, that the approach was able to obtain better results than other techniques used in the literature. The techniques in the literature that obtained the best results were those based on Bayesian networks, surpassing other techniques such as Markov, ad hoc probabilistic predictors, decision trees, SVM, among others. By combining different predictors, the approach proposed in this work surpassed Bayesian networks, achieving, on average, 5.3 % more accuracy for all datasets used. In turn, it was shown that datasets containing short stays (less than 10 minutes) are more difficult to predict than datasets that only have long stays (8.2 % better accuracy for Nokia-20 dataset w.r.t Nokia-10 dataset).

The approach presented has some limitations that will be addressed in future work. First, the aging mechanism employed may take some time to identify modifications in users' routines. This may not be a prob-

[1] All experiments were run on Apple M1 and 16GB of RAM

lem if there are minor changes in the user's routine, but if the change is significant, it can have a high impact on the accuracy of predictions. In particular, two possible situations are identified that should be addressed in a manner that is independent of the aging mechanism currently used: the user changing their home or place of work or the user taking a vacation without traveling outside of their city. A change of home or work will have a strong impact on predictions, as these are the places the user frequents most. The current approach will take some time to forget all the routines involving the old home or workplace and then learn the routines with the new places. On the other hand, if the user is on vacation, they will abruptly modify their routines, and upon finishing their vacation, they will return to their daily routines. It is likely that the vacation will not last long enough for the approach to learn the user's routines during their vacation. Second, the approach predicts the user's next destination only when it detects that the user has left the place where he/she was. Some services may require knowing the user's next destination before they start their trip. At this point, it is necessary to predict when the user's current stay will end, so that their next destination can be predicted before starting the trip.

Finally, the last limitation identified is the difficulty of building a dataset that records in detail the daily mobility of citizens. In this work, a dataset was created to carry out experimental evaluations. This dataset has detailed records of the daily mobility of a set of volunteer users for periods of up to more than a year. During this period, users used a prototype of the proposed approach that collected data on the user's location and physical activity and processed it to detect the user's stays, trips, and places visited. The application asks the volunteers for information whenever it was required, and allows them to edit or delete visits or places that they consider erroneous. However, there are still some shortcomings when building the dataset. First, it is difficult to know the exact moment when the user enters or leaves a place. Asking the user to specify when they entered and left each place would not only be very time-consuming, but it is also likely that the user themselves do not know the exact moment at which a stay begins or ends. For this, the use of predefined visit tours seem to be the best option. Second, disseminating the constructed datasets is difficult and controversial, as it seriously compromises the privacy of users. At this point, it is necessary to find a data anonymization mechanism that allows protecting the privacy of the data without hindering the use of the dataset for other researchers.

6. Conclusions

We proposed a prediction approach that combines multiple mobility predictors through an adaptive weighting mechanism tailored to each user. The method uses nine probabilistic-matrix-based predictors, each capturing different aspects of user mobility, and dynamically prioritizes the most accurate ones. Experimental results show that our approach outperforms Markov-based models by 13.1 %, decision trees by 8.1 %, and Bayesian networks by 5.3 % in accuracy, while maintaining a low resource usage of only 48 MB of RAM on average. This makes it suitable for deployment on low-power mobile devices.

Some limitations of the proposal include: the aging mechanism make take some time to detect changes in users' routines, such as when the user moves to another town; the prediction of the next location occurs after the user have left the previous one, which might be not useful for some services that require to know the next place before the user leaves the current one. As a future work we plan to address these issues as well as we plan to consider the problem of urban mobility in the context of groups or communities.

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Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Sebastian Vallejos: Conceptualization, Methodology, Software, Validation, Data curation; **Luis S. Berdun:** Conceptualization, Methodology, Validation, Data curation, Writing - review & editing, Supervision, Funding acquisition; **Ariel J. Monteserin:** Software, Validation, Writing - review & editing, Funding acquisition; **Daniela L. Godoy:** Validation, Writing - review & editing, Funding acquisition; **Marcelo G. Armentano:** Methodology, Data curation, Writing - original draft, Writing - review & editing, Supervision, Funding acquisition; **Silvia N. Schiaffino:** Validation, Writing - original draft, Writing - review & editing, Supervision, Funding acquisition.

References

Alvarez-Garcia, J. A., Ortega, J. A., Gonzalez-Abril, L., & Velasco, F. (2010). Trip destination prediction based on past GPS log using a hidden markov model. *Expert Systems with Applications*, *37*(12), 8166–8171. https://doi.org/10.1016/j.eswa.2010.05.070

Ashbrook, D., & Starner, T. (2003). Using GPS to learn significant locations and predict movement across multiple users. *Personal and Ubiquitous Computing*, *7*(5), 275–286. https://doi.org/10.1007/S00779-003-0240-0

Chekol, A. G., & Fufa, M. S. (2022). A survey on next location prediction techniques, applications, and challenges. *EURASIP Journal on Wireless Communications and Networking*, *29*. https://doi.org/10.1186/s13638-022-02114-6

Comito, C. (2020). Next: A framework for next-place prediction on location based social networks. *Knowledge-Based Systems*, *204*, 106–205. https://doi.org/10.1016/j.knosys.2020.106205

Dash, M., Koo, K. K., Gomes, J. B., Krishnaswamy, S. P., Rugeles, D., & Shi-Nash, A. (2015). Next place prediction by understanding mobility patterns. In *2015 IEEE International conference on pervasive computing and communication workshops (percom workshops)* (pp. 469–474). https://doi.org/10.1109/PERCOMW.2015.7134083

Etter, V., Kafsi, M., Kazemi, E., Grossglauser, M., & Thiran, P. (2013). Where to go from here? Mobility prediction from instantaneous information. *Pervasive and Mobile Computing*, *9*(6), 784–797. Mobile Data Challenge. https://doi.org/10.1016/j.pmcj.2013.07.006

Feng, J., Li, Y., Zhang, C., Sun, F., Meng, F., Guo, A., & Jin, D. (2018). Deepmove: Predicting human mobility with attentional recurrent networks. In *Proceedings of the 2018 world wide web conference* WWW '18 (p. 1459-1468). Republic and Canton of Geneva, CHE: International World Wide Web Conferences Steering Committee. https://doi.org/10.1145/3178876.3186058

Gambs, S., Killijian, M.-O., & del Prado Cortez, M. N. n. (2012). Next place prediction using mobility markov chains. In *Proceedings of the first workshop on measurement, privacy, and mobility (MPM '12)*. Bern, Switzerland: Association for Computing Machinery. https://doi.org/10.1145/2181196.2181199

Gao, H., Tang, J., & Liu, H. (2012). Mobile location prediction in spatio-temporal context. In *Nokia mobile data challenge workshop* (pp. 1–4). (*vol. 41*).

Garola, G., Siragusa, C., Seghezzi, A., & Mangiaracina, R. (2024). Next place prediction model: A literature review. https://doi.org/10.3233/ATDE240018

Gomes, J. B., Phua, C., & Krishnaswamy, S. (2013). Where will you go? Mobile data mining for next place prediction. In L. Bellatreche, & M. K. Mohania (Eds.), *Data warehousing and knowledge discovery* (pp. 146–158). Springer Berlin Heidelberg.

Herder, E., Siehndel, P., & Kawase, R. (2014). Predicting user locations and trajectories. In V. Dimitrova, T. Kuflik, D. Chin, F. Ricci, P. Dolog, & G.-J. Houben (Eds.), *User modeling, adaptation, and personalization* (pp. 86–97). Springer.

Imai, R., Tsubouchi, K., Konishi, T., & Shimosaka, M. (2018). Early destination prediction with spatio-temporal user behavior patterns. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *1*(4). https://doi.org/10.1145/3161197

Kim, B., Ha, J.-Y., Lee, S., Kang, S., Lee, Y., Rhee, Y., Nachman, L., & Song, J. (2011). ADNext: A visit-pattern-aware mobile advertising system for urban commercial complexes. In *Proceedings of the 12th workshop on mobile computing systems and applications* HotMobile '11 (p. 7-12). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/2184489.2184492

Kulkarni, V., Moro, A., & Garbinato, B. (2016). Mobidict: A mobility prediction system leveraging realtime location data streams. In *Proceedings of the 7th ACM SIGSPATIAL international workshop on geostreaming (IWGS '16)*. Burlingame, California: Association for Computing Machinery. https://doi.org/10.1145/3003421.3003424

Laurila, J. K., Gatica-Perez, D., Aad, I., Blom, J., Bornet, O., Do, T. M. T., Dousse, O., Eberle, J., & Miettinen, M. (2013). From big smartphone data to worldwide research: The mobile data challenge. *Pervasive and Mobile Computing*, *9*(6), 752–771. Mobile Data Challenge. https://doi.org/10.1016/j.pmcj.2013.07.014

Liu, J., Chen, Y., Huang, X., Li, J., & Min, G. (2023). GNN-based long and short term preference modeling for next-location prediction. *Information Sciences*, *629*, 1–14. https://doi.org/10.1016/j.ins.2023.01.131

Liu, Q., Wu, S., Wang, L., & Tan, T. (2016). Predicting the next location: A recurrent model with spatial and temporal contexts. In *Proceedings of the thirtieth AAAI conference on artificial intelligence* AAAI'16 (p. 194-200). AAAI Press.

Ma, Z., & Zhang, P. (2022). Individual mobility prediction review: Data, problem, method and application. *Multimodal Transportation*, *1*(1), 100002. https://doi.org/10.1016/j.multra.2022.100002

Manning, C. D., Raghavan, P., & Schütze, H. (2008). Introduction to information retrieval. New York, NY, USA: Cambridge University Press. https://doi.org/10.1017/CBO9780511809071

Nguyen, T., Cheng, H.-T., Wu, P., Buthpitiya, S., Zhu, J., & Zhang, Y. (2012). PnLUM : System for prediction of next location for users with mobility. In *Nokia mobile data challenge workshop*. (*vol. 2*).

Nishino, M., Nakamura, Y., Yagi, T., Muto, S., & Abe, M. (2010). A location predictor based on dependencies between multiple lifelog data. In *Proceedings of the 2nd ACM SIGSPATIAL international workshop on location based social networks (LBSN '10)* (p. 11-17). San Jose, California: Association for Computing Machinery. https://doi.org/10.1145/1867699.1867702

Prabhala, B., & La Porta, T. (2015). Spatial and temporal considerations in next place predictions. In *2015 IEEE Conference on computer communications workshops (INFOCOM WKSHPS)* (pp. 390–395). https://doi.org/10.1109/INFCOMW.2015.7179416

Prabhala, B., Wang, J., Deb, B., La Porta, T., & Han, J. (2014). Leveraging periodicity in human mobility for next place prediction. In *2014 IEEE Wireless communications and networking conference (WCNC)* (pp. 2665–2670). https://doi.org/10.1109/WCNC.2014.6952829

Qin, Z., Zhang, P., & Ma, Z. (2024). DeepAGS: Deep learning with activity, geography and sequential information in predicting an individual's next trip destination. *IET Intelligent Transport Systems*, *18*, 1895–1909. https://doi.org/10.1049/itr2.12554

Shams, B., & Haratizadeh, S. (2018). GraphLoc: A graph based approach for automatic detection of significant locations from GPS trajectory data. *Journal of Spatial Science*, *63*(1), 115–134. https://doi.org/10.1080/14498596.2017.1327374

Solomon, A., Livne, A., Katz, G., Shapira, B., & Rokach, L. (2021). Analyzing movement predictability using human attributes and behavioral patterns. *Computers, Environment and Urban Systems*, *87*, 101596. https://doi.org/10.1016/j.compenvurbsys.2021.101596

Song, L., Kotz, D., Jain, R., & He, X. (2006). Evaluating next-cell predictors with extensive wi-fi mobility data. *IEEE Transactions on Mobile Computing*, *5*(12), 1633–1649. https://doi.org/10.1109/TMC.2006.185

Sun, J., & Kim, J. (2021). Joint prediction of next location and travel time from urban vehicle trajectories using long short-term memory neural networks. *Transportation Research Part C: Emerging Technologies*, *128*, 103–114. https://doi.org/10.1016/j.trc.2021.103114

Sun, K., Qian, T., Chen, T., Liang, Y., Nguyen, Q. V. H., & Yin, H. (2020). Where to go next: Modeling long- and short-term user preferences for point-of-interest recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(01), 214–221. https://doi.org/10.1609/aaai.v34i01.5353

Vallejos, S., Berdún, L., Armentano, M. G., Schiaffino, S. N., & Godoy, D. (2024). A lightweight approach for building user mobility profiles. *ISPRS International Journal of Geo-Information*, *13*, 11. https://api.semanticscholar.org/CorpusID:267431383.

Wilinski, A. (2019). Time series modeling and forecasting based on a markov chain with changing transition matrices. *Expert Systems with Applications*, *133*, 163–172. https://doi.org/10.1016/j.eswa.2019.04.067

Wu, H., Chen, Z., Sun, W., Zheng, B., & Wang, W. (2017). Modeling trajectories with recurrent neural networks. In *Proceedings of the twenty-sixth international joint conference on artificial intelligence, IJCAI-17* (pp. 3083–3090). https://doi.org/10.24963/ijcai.2017/430

Yan, M., Li, S., Chan, C. A., Shen, Y., & Yu, Y. (2021). Mobility prediction using a weighted markov model based on mobile user classification. *Sensors*, *21*(5). https://doi.org/10.3390/s21051740

Yanagihara, T., Namiki, R., Nawa, K., Weir, D., & Oguchi, K. (2013). Combining prediction methods with cyber information for proactive route recommendation. In *2013 IEEE International conference on cyber technology in automation, control and intelligent systems* (pp. 87–91). https://doi.org/10.1109/CYBER.2013.6705425

Yang, J., Xu, J., Xu, M., Zheng, N., & Chen, Y. (2014). Predicting next location using a variable order markov model. In *Proceedings of the 5th ACM SIGSPATIAL international workshop on geostreaming* IWGS '14 (p. 37-42). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/2676552.2676557

Yao, Y., Guo, Z., Dou, C., Jia, M., Hong, Y., Guan, Q., & Luo, P. (2023). Predicting mobile users' next location using the semantically enriched geo-embedding model and the multilayer attention mechanism. *Computers, Environment and Urban Systems*, *104*, 102009. https://doi.org/10.1016/j.compenvurbsys.2023.102009

Yi, F., Li, Z., Wang, H., Zheng, W., & Sun, L. (2017). MIAC: A mobility intention autocompletion model for location prediction. In G. Li, Y. Ge, Z. Zhang, Z. Jin, & M. Blumenstein (Eds.), *Knowledge science, engineering and management* (pp. 432–444). Springer International Publishing.

Yu, C., Liu, Y., Yao, D., Yang, L. T., Jin, H., Chen, H., & Ding, Q. (2017). Modeling user activity patterns for next-place prediction. *IEEE Systems Journal*, *11*(2), 1060–1071. https://doi.org/10.1109/JSYST.2015.2445919

Yu, P., Zhang, X., Gong, Y., Zhang, J., Sun, H., Zhang, J., Zhang, X., & Yin, Y. (2025). Enhancing origin-destination flow prediction via bi-directional spatio-temporal inference and interconnected feature evolution. *Expert Systems with Application*, *264*(C). https://doi.org/10.1016/j.eswa.2024.125679

Zeng, J., Zhao, Y., Wang, Z., Tao, H., Gao, M., & Wen, J. (2023). LGSA: A next poi prediction method by using local and global interest with spatiotemporal awareness. *Expert Systems with Applications*, *227*, 120291. https://doi.org/10.1016/j.eswa.2023.120291

Zhang, P., Koutsopoulos, H. N., & Ma, Z. (2023). Deeptrip: A deep learning model for the individual next trip prediction with arbitrary prediction times. *IEEE Transactions on Intelligent Transportation Systems*, *24*(6), 5842–5855. https://doi.org/10.1109/TITS.2023.3252043

Zhu, Y., Sun, Y., & Wang, Y. (2012). Nokia mobile data challenge: Predicting semantic place and next place via mobile data. In *Nokia mobile data challenge workshop*.

Zolotukhin, M., Ivannikova, E., & Hämäläinen, T. (2013). Novel method for the prediction of mobile location based on temporal-spatial behavioral patterns. In *2013 IEEE Third international conference on information science and technology (ICIST)* (pp. 761–766). https://doi.org/10.1109/ICIST.2013.6747655

Zong, F., Tian, Y., He, Y., Tang, J., & Lv, J. (2019). Trip destination prediction based on multi-day GPS data. *Physica A: Statistical Mechanics and its Applications*, *515*, 258–269. https://doi.org/10.1016/j.physa.2018.09.090